Exercise 3D: Complete Solutions

TEXTBOOK METHODOLOGY & RULES FOR DIVISION

1. **Dividing a Decimal by a Power of 10** (10, 100, 1000, etc.): Shift the decimal point of the dividend to the **left** by as many decimal places as there are zeros in the divisor.

2. **Dividing a Decimal by a Whole Number**: Perform division as with whole numbers first. Place the decimal point in the quotient directly above/when the division of the whole-number part of the dividend is completed.

3. **Dividing a Decimal by a Decimal**: Convert the divisor into a whole number by multiplying both dividend and divisor by a suitable power of 10 (10, 100, 1000, etc.). Then, divide by the new whole number.

1. Division by Powers of 10 (10, 100, 1000)

For each sub-question below, division is executed by moving the decimal point leftwards according to the number of zeros in the divisor. Zeros are annexed where necessary to serve as placeholders.

Part	Expression	Zeros in Divisor	Decimal Shift & Method	Final Quotient
(i)	2.8 ÷ 10	1 Zero	Shift left by 1 place: .28	**0.28**
(ii)	0.63 ÷ 10	1 Zero	Shift left by 1 place: .063	**0.063**
(iii)	3.05 ÷ 10	1 Zero	Shift left by 1 place: .305	**0.305**
(iv)	245.6 ÷ 100	2 Zeros	Shift left by 2 places: 2.456	**2.456**
(v)	0.9 ÷ 100	2 Zeros	Shift left by 2 places: .009	**0.009**
(vi)	1.23 ÷ 100	2 Zeros	Shift left by 2 places: .0123	**0.0123**
(vii)	134.2 ÷ 1000	3 Zeros	Shift left by 3 places: .1342	**0.1342**
(viii)	23.4 ÷ 1000	3 Zeros	Shift left by 3 places: .0234	**0.0234**
(ix)	0.7 ÷ 1000	3 Zeros	Shift left by 3 places: .0007	**0.0007**

2. Division of Decimals by Whole Numbers

Divide as with integers, then place the decimal point in the quotient directly aligned with the dividend's point.

Part	Expression	Whole Number Operation	Decimal Placement	Final Quotient
(i)	20.79 ÷ 9	2079 ÷ 9 = 231	Set 2 decimal places: 2.31	**2.31**
(ii)	78.48 ÷ 12	7848 ÷ 12 = 654	Set 2 decimal places: 6.54	**6.54**
(iii)	142.8 ÷ 21	1428 ÷ 21 = 68	Set 1 decimal place: 6.8	**6.8**
(iv)	6.02 ÷ 7	602 ÷ 7 = 86	Set 2 decimal places: 0.86	**0.86**
(v)	0.688 ÷ 8	688 ÷ 8 = 86	Set 3 decimal places: 0.086	**0.086**
(vi)	0.125 ÷ 25	125 ÷ 25 = 5	Set 3 decimal places: 0.005	**0.005**
(vii)	0.992 ÷ 31	992 ÷ 31 = 32	Set 3 decimal places: 0.032	**0.032**
(viii)	0.1728 ÷ 72	1728 ÷ 72 = 24	Set 4 decimal places: 0.0024	**0.0024**
(ix)	0.12749 ÷ 61	12749 ÷ 61 = 209	Set 5 decimal places: 0.00209	**0.00209**

3. Division of Decimals by Decimals

Multiply both terms by the power of 10 that makes the divisor a whole number, then divide.

Part	Expression	Power of 10	Equivalent Whole Number Divisor Equation	Final Quotient
(i)	2.24 ÷ 0.8	× 10	22.4 ÷ 8 = 2.8	**2.8**
(ii)	1.242 ÷ 1.8	× 10	12.42 ÷ 18 = 0.69	**0.69**
(iii)	0.1575 ÷ 0.21	× 100	15.75 ÷ 21 = 0.75	**0.75**
(iv)	0.0144 ÷ 0.12	× 100	1.44 ÷ 12 = 0.12	**0.12**
(v)	0.0783 ÷ 0.9	× 10	0.783 ÷ 9 = 0.087	**0.087**
(vi)	0.1164 ÷ 0.012	× 1000	116.4 ÷ 12 = 9.7	**9.7**
(vii)	0.068 ÷ 0.17	× 100	6.8 ÷ 17 = 0.4	**0.4**
(viii)	0.02324 ÷ 0.28	× 100	2.324 ÷ 28 = 0.083	**0.083**
(ix)	0.03822 ÷ 0.049	× 1000	38.22 ÷ 49 = 0.78	**0.78**

4. Simplifying Combined Expressions

Solve numerators and denominators independently first, then carry out the final division.

Part	Expression	Numerator Simplification	Denominator Simplification	Final Answer
(i)	(1.3 × 2.4) ÷ 0.39	1.3 × 2.4 = 3.12	0.39	**8**
(ii)	(2.5 × 40.4) ÷ 50	2.5 × 40.4 = 101.0	50	**2.02**
(iii)	6.3 ÷ (0.3 × 0.1)	6.3	0.3 × 0.1 = 0.03	**210**

Practical Applications & Word Problems

Problem 5

Question: A boy bought 8 pencils for Rs. 46.80. What is the cost of each pencil?

Given Information:
Total number of pencils = 8
Total cost of pencils = Rs. 46.80

Step-by-Step Working:
- Let the cost of one pencil be x.
- Using the unitary method, Cost of 1 pencil = Total Cost ÷ Number of Pencils.
- Calculate: 46.80 ÷ 8
- Divide 4680 by 8 to get 585. Since there are 2 decimal places, set the point: 5.85

Final Answer: **Rs. 5.85 per pencil**

Problem 6

Question: A car covers a distance of 276.75 km in 4.5 hours. What is the average speed of the car?

Given Information:
Total distance covered = 276.75 km
Total time taken = 4.5 hours

Step-by-Step Working:
- Formula: Average Speed = Total Distance ÷ Total Time.
- Write the equation: Speed = 276.75 ÷ 4.5
- Multiply both numbers by 10 to clear the divisor's decimal: 2767.5 ÷ 45
- Perform long division: 27675 ÷ 45 = 615. Place the decimal point: 61.5

Final Answer: **61.5 km/hour**

Problem 7

Question:	2.25 m of a cloth costs Rs. 326.25. What is the cost of 1 m of cloth?
Given Information:	Length of cloth = 2.25 meters Total cost of cloth = Rs. 326.25
Step-by-Step Working:	• Unitary Method: Cost of 1 meter of cloth = Total Cost ÷ Length of cloth. • Write the equation: Cost of 1 m = 326.25 ÷ 2.25 • Multiply both by 100 to make the divisor a whole number: 32625 ÷ 225 • Perform long division: 32625 ÷ 225 = 145
Final Answer:	**Rs. 145.00 per meter**

Problem 8

Question:	The product of two numbers is 52.8. If one of them is 8.25, find the other.
Given Information:	Product of numbers = 52.8 One number = 8.25
Step-by-Step Working:	• Let the other number be x. • Equation: 8.25 × x = 52.8 ==> x = 52.8 ÷ 8.25 • Multiply both terms by 100 to clear the divisor's decimal: 5280 ÷ 825 • Perform long division: 5280 ÷ 825 = 6.4
Final Answer:	**6.4**

Problem 9

Question:	How many equal pieces, each of length 3.6 cm can be cut from a rope of length 61.2 cm?
Given Information:	Total length of rope = 61.2 cm Length of each piece = 3.6 cm
Step-by-Step Working:	• Let the number of pieces be n. • Equation: n = Total Length ÷ Piece Length = 61.2 ÷ 3.6 • Multiply both terms by 10 to eliminate decimals: 612 ÷ 36 • Perform division: 612 ÷ 36 = 17
Final Answer:	**17 equal pieces**